

Predictable first-chaos origin force and anchor-free semiconvexity reduction

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope

R-141 identifies the legal finite-chart projected force of the complete source, covariance-normal endpoint, and terminal sixth-power action. R-146 then proves that A7 centering fixes the zero-control additive gauge, so the direct endpoint identity is exactly relative and needs no separate absolute anchor. R-163 nevertheless retained both an anchor and an origin-force bound as hypotheses of its general reduced-action semiconvexity criterion.

This note closes the origin-force hypothesis on the direct zero-control Gaussian chart. It also gives a force-only pullback corollary for uniformly bounded origin-preserving causal graphs; the full T-050 composition extends to such a graph only if its source metric and stabilizer decomposition are separately transported without change. The mechanism is not pointwise force regularity. It is the orthogonal projection of a terminal scalar onto the predictable first Gaussian chaos. For every finite temporally faithful chart,

$$\sum_b \mathbb{E} \|g_b(0)\|^2 \leq \text{Var}(F_J), \quad (1.1)$$

where F_J is the complete zero-control endpoint scalar. A7 gives a cutoff-uniform L^2 bound for the renormalized Class-II energy, and the A1/A11 point-covariance trace gives a uniform twelfth Gaussian moment for the sixth-power stabilizer. Hence the right side of (1.1) is uniform in cutoff, chart length, and independent-Gaussian linear/isometric refinement preserving the same zero-control law.

Combining this with the exact R-146 zero-control anchor sharpens the R-163 criterion. If, for one $\rho > 0$, the complete direct owner satisfies

$$K_{\text{owner}} \succeq - \left(\frac{10}{11} - \rho \right) G_{\text{src}} \quad (1.2)$$

on every radial segment, then

$$\mathbb{E} V_J^{\text{ren}}(Z_h) \geq - \max \left\{ 0, \frac{5}{11} - \frac{\rho}{4} \right\} X(h) - \frac{3}{20} Y_6(h) - \frac{B_*}{\rho}. \quad (1.3)$$

Thus the direct route, corresponding to $C_\chi = 1$, no longer needs separate anchor or origin-force assumptions. Equation (1.2), including forward, legal reverse, balanced, low, trace, current, future, forest, and sextic connections exactly once, remains unproved in production scope. Therefore T-050, A13, Nelson, removal, the interacting measure, every phase/PDE question, and Sector A remain open.

The result is deliberately not asserted for stationary low elimination whose base minimizer is not zero, for a graph which changes the zero-control law, or for a nonzero adapted feedback base. In those cases the Malliavin derivative contains feedback Jacobians and the A7 base variance does not control the required fixed-law legal force.

2. Predictable first-Gaussian-chaos projection

Let $E_1, \dots, E_B$ be finite-dimensional real Hilbert spaces and let $\xi_b \sim N(0, \mathbf{I}_{E_b})$ be independent. Put

$$\mathcal{F}_b = \sigma(\xi_1, \dots, \xi_b), \quad \mathcal{F}_0 = \{\emptyset, \Omega\}. \quad (2.1)$$

For a smooth $F \in L^2$ with square-integrable Gaussian derivatives, define

$$g_b = \mathbb{E}[\nabla_b F \mid \mathcal{F}_{b-1}]. \quad (2.2)$$

The predictable source space is

$$\mathcal{H}_{\text{pred}} = \bigoplus_{b=1}^B L^2(\Omega, \mathcal{F}_{b-1}; E_b), \quad \|k\|_{\text{pred}}^2 = \sum_b \mathbb{E}\|k_b\|^2. \quad (2.3)$$

Define its adapted first-chaos integral by

$$\delta k = \sum_b \langle k_b, \xi_b \rangle. \quad (2.4)$$

Theorem 2.1 (sharp predictable first-chaos bound)

The map $\delta : \mathcal{H}_{\text{pred}} \rightarrow L^2(\Omega)$ is an isometry and

$$\boxed{\sum_b \mathbb{E}\|g_b\|^2 \leq \text{Var}(F).} \quad (2.5)$$

More precisely, δg is the orthogonal projection of $F - \mathbb{E}F$ onto the closed range of δ . The constant one is sharp.

Proof. For $b < c$, conditioning on $\mathcal{F}_{c-1}$ and then on the fresh centered block ξ_c makes the b, c cross term in $\mathbb{E}(\delta k)^2$ vanish. The diagonal terms give

$$\mathbb{E}(\delta k)^2 = \sum_b \mathbb{E}\|k_b\|^2. \quad (2.6)$$

For predictable k_b , Gaussian integration by parts in the fresh block ξ_b has no derivative of k_b , and therefore

$$\mathbb{E}[(F - \mathbb{E}F)\delta k] = \sum_b \mathbb{E}\langle \nabla_b F, k_b \rangle = \sum_b \mathbb{E}\langle g_b, k_b \rangle = \mathbb{E}[(\delta g)(\delta k)]. \quad (2.7)$$

Thus $F - \mathbb{E}F - \delta g$ is orthogonal to the range of δ . Pythagoras and (2.6) prove (2.5). Taking $F = \langle u, \xi_1 \rangle$ gives equality, so the constant is sharp. $\square$

Equivalently, if $M_b = \mathbb{E}[F \mid \mathcal{F}_b]$, conditional Gaussian integration by parts and Bessel's inequality in the current block give

$$g_b = \mathbb{E}[\xi_b(M_b - M_{b-1}) \mid \mathcal{F}_{b-1}], \quad \|g_b\|^2 \leq \mathbb{E}[(M_b - M_{b-1})^2 \mid \mathcal{F}_{b-1}]. \quad (2.8)$$

Summation uses the orthogonality of the martingale increments. This second proof makes clear why a discrete chart generally gives an inequality. In a completed Brownian filtration with the predictable-representation property, the full Clark–Ocone integral gives equality.

3. Identification with the R-141 origin force

Fix a finite cutoff J and a temporally faithful zero-control chart with deterministic R-104 synthesis operators

$$Z_0 = \sum_{b=1}^B S_b \xi_b \stackrel{\text{law}}{=} X_J. \quad (3.1)$$

The R-104 terminal-stage identity is pathwise: $Z_{K,0} = Z_0 = S_\pi \xi$. Define the terminal functional and its zero-control random scalar by

$$\mathfrak{G}_J(\phi) = V_J^{\text{ren}}(\phi) + \frac{3}{20} \|\phi\|_6^6, \quad F_J := \mathfrak{G}_J(Z_0). \quad (3.2)$$

At finite cutoff this is a smooth finite-dimensional scalar. The R-141 formula and the differentiated R-128 owner identity give

$$(g_{J,\pi}^{\text{prod}}(0))_b = \mathbb{E}[S_b^* D \mathfrak{G}_J(Z_0) \mid \mathcal{F}_{b-1}] = \mathbb{E}[\nabla_b F_J \mid \mathcal{F}_{b-1}]. \quad (3.3)$$

The derivative of the explicit $9\|h\|^2/20$ source cost is zero at the origin. Theorem 2.1 therefore gives the chart-uniform statement

$$\boxed{\|g_{J,\pi}^{\text{prod}}(0)\|_{\text{src},*}^2 \leq \text{Var}(F_J)}. \quad (3.4)$$

No field/current pointwise norm and no inverse synthesis occurs in (3.4). The estimate is independent of the number of source blocks and survives every independent-Gaussian linear/isometric source refinement that preserves the zero-control terminal law, because each refined gradient is still an orthogonal projection of the same centered endpoint scalar.

The force in (3.3) need not vanish. A7 centering states $\mathbb{E}V_J^{\text{ren}}(X_J) = 0$, which is a value normalization. R-153 gives the strict-past first-variation formula and shows that the registered structure does not force it to vanish. Thus neither centering nor spatial stationarity may be substituted for (3.4).

Corollary 3.1 (origin-preserving causal graphs)

Let χ be an admissible reduced or graph chart with $\chi(0) = 0$, the same zero-control terminal law, and a causal tangent $R_0 = D\chi(0)$ satisfying

$$\|R_0\|_{\mathcal{H}_{\text{red}} \rightarrow \mathcal{H}_{\text{pred}}} \leq C_\chi. \quad (3.5)$$

Assume that the same constant C_χ works uniformly over the declared family of graphs. Then

$$\|D(\mathcal{A} \circ \chi)(0)\|_{\text{red},*}^2 \leq C_\chi^2 \text{Var}(F_J). \quad (3.6)$$

This corollary includes a prescribed nonlinear graph only at first order; its Hessian still contains the registered projected-force connection $D\mathcal{A}(0)D^2\chi(0)$. Changing the graph metric is not a free normalization: it also changes the source cost and the owner threshold. Therefore (3.6) alone does not transfer Theorem 5.1 to an arbitrary graph.

4. Cutoff-uniform endpoint variance and exact anchor

A7 proves

$$\mathcal{V}_J^{\text{ren}} := \int e_{\Pi,J}^{\text{ren}}(x) \, dx \longrightarrow \mathcal{V}^{\text{ren}} \quad \text{in } L^2(\Omega). \quad (4.1)$$

Consequently

$$M_7 := \sup_J \|\mathcal{V}_J^{\text{ren}}\|_{L^2} < \infty. \quad (4.2)$$

The A1 symbol has order minus four in three dimensions, and A11 records the corresponding point-covariance trace convergence. On the fixed side-16, positive-floor, registered sharp-cube/dyadic cutoff sequence put

$$T_* := \sup_{J,x} \mathbb{E}|Z_{K,0}(x)|^2 < \infty. \quad (4.3)$$

Here K is the registered terminal stage: the R-104 zero-control law gives $Z_{K,0} \stackrel{\text{law}}{=} X_J$. Thus (4.3) is the same all-cutoff trace ceiling, not an unproved contraction estimate.

For a centered finite-dimensional real Gaussian G with covariance Σ , weighted Jensen gives

$$\mathbb{E}|G|^{12} \leq 11!! (\text{Tr } \Sigma)^6. \quad (4.4)$$

If $\text{Tr } \Sigma = 0$, then $G = 0$ almost surely. Otherwise, write $|G|^2 = \sum_a \lambda_a g_a^2$ and apply convexity to the weights $\lambda_a / \text{Tr } \Sigma$, followed by $\mathbb{E}g^{12} = 11!!$. This is the safe sharp constant for the general real Gaussian class; the circular-complex production subclass may be sharper. With $|\mathbb{T}_L^3| = L^3$, spatial Cauchy–Schwarz therefore yields

$$\left\| \|Z_{K,0}\|_6^6 \right\|_{L^2} \leq L^3 \sqrt{11!!} T_*^3 =: M_6. \quad (4.5)$$

Minkowski and $\text{Var } F \leq \mathbb{E}|F|^2$ now give the explicit uniform certificate

$$\boxed{\sup_{J,\pi} \|g_{J,\pi}^{\text{prod}}(0)\|_{\text{src},*}^2 \leq B_* := \left(M_7 + \frac{3}{20}M_6\right)^2 < \infty.} \quad (4.6)$$

The theorem uses the analytic all-cutoff quantity T_* on that registered cutoff sequence. A finite-ladder numerical trace ceiling may audit (4.6), but is not substituted for (4.3).

The exact zero-control anchor is simultaneously favourable. A7 centering and (3.2) give

$$\mathcal{A}_{J,\pi}(0) = \mathbb{E}\mathcal{V}_J^{\text{ren}} + \frac{3}{20}\mathbb{E}\|Z_{K,0}\|_6^6 = \frac{3}{20}\mathbb{E}\|Z_{K,0}\|_6^6 \geq 0. \quad (4.7)$$

This is the direct-action version of R-146 Theorem 2.1. No separate cutoff-uniform bound on the representation-dependent constants $a_{J,\pi}$ and $\mathcal{T}_{J,\pi}(0)$ is inferred.

5. Anchor-free T-050 semiconvexity reduction

Let $X(h) = \|h\|_{\text{src}}^2$ and

$$\mathcal{A}_{J,\pi}(h) = \mathbb{E}V_J^{\text{ren}}(Z_h) + \frac{9}{20}X(h) + \frac{3}{20}Y_6(h) \quad (5.1)$$

be the direct complete action. Assume one uniform $\rho > 0$ and, on every radial segment,

$$D^2\mathcal{A}_{J,\pi}(th)[h, h] \geq \left(-\frac{1}{110} + \rho\right) X(h), \quad 0 \leq t \leq 1. \quad (5.2)$$

This is exactly what follows from the explicit $9G_{\text{src}}/10$ source Hessian and the owner condition (1.2), where the notation is fixed by

$$D^2\mathcal{A} = \frac{9}{10}G_{\text{src}} + K_{\text{owner}}. \quad (5.2a)$$

Thus the source Hessian is counted once, not included again inside K_{owner} .

Theorem 5.1 (one remaining complete-owner gate)

Under (3.1)–(4.3) and (5.2),

$$\boxed{\mathbb{E}V_J^{\text{ren}}(Z_h) \geq -\epsilon_v X(h) - \frac{3}{20}Y_6(h) - \frac{B_*}{\rho}, \quad \epsilon_v = \max\left\{0, \frac{5}{11} - \frac{\rho}{4}\right\} < \frac{5}{11}.} \quad (5.3)$$

Since $3/20 < 27/100$, (5.3) is the registered T-050 inequality on every direct chart for which the complete owner hypothesis holds uniformly.

Proof. Taylor’s formula, (4.7), and (3.4) give

$$\mathcal{A}(h) \geq -\sqrt{B_*X(h)} + \frac{1}{2}\left(-\frac{1}{110} + \rho\right)X(h). \quad (5.4)$$

Use Young’s inequality with $\eta = \rho/4$:

$$-\sqrt{B_*X} \geq -\frac{\rho}{4}X - \frac{B_*}{\rho}. \quad (5.5)$$

Subtracting the two stabilizers in (5.1) leaves the source coefficient

$$\frac{9}{20} - \frac{1}{2}\left(-\frac{1}{110} + \rho\right) + \frac{\rho}{4} = \frac{5}{11} - \frac{\rho}{4}. \quad (5.6)$$

If it is negative, discard the favourable multiple of X . This proves (5.3). $\square$

Theorem 5.1 does not supply (5.2). It says that for the direct route, a uniform complete-owner lower bound above $-(10/11)G_{\text{src}}$ is now the sole new analytic inequality in the R-163 semiconvexity implication. Every block must still be assembled in one self-adjoint owner: forward and legal reverse are one adjoint pair, while balanced and low are other blocks, not new reserves.

For a graph, the same calculation gives constant $C_\chi^2 B_*/\rho$ only after one separately proves that the pulled-back action still has the exact form (5.1) in one fixed reduced source metric and that (5.2) holds in that same metric. Corollary 3.1 alone is not that proof.

6. Scope firewalls and failed shortcuts

1. **Centering does not make the force zero.** R-146 subtracts a constant. It removes the absolute anchor but leaves the first variation unchanged. R-153 records a predictable strict-past first-variation formula that is not identically forced to vanish.
2. **Theorem 2.1 is an origin theorem.** For nonzero feedback h ,

$$D_{\xi_b}Z_h = S_b + \sum_{c>b} S_c D_b h_c. \quad (6.1)$$

Clark–Ocone then controls the feedback-corrected Malliavin force, not the R-141 fixed-law legal control force. No uniform Malliavin derivative or inverse feedback Jacobian is part of the Boue–Dupuis source budget.

3. **A stationary low base may move the origin.** If independent-low elimination produces $\ell_*(0) \neq 0$, then the reduced base endpoint is not the A7 zero-control Gaussian. Equations (3.4), (4.6), and (4.7) do not automatically transfer. One must instead prove a uniform shifted endpoint variance and the exact reduced tangent, or use the full direct action.
4. **Scalar L^2 convergence is used only at the base.** A7 does not imply a pointwise or nonzero-feedback force norm. The predictable projection theorem converts the zero-control scalar variance to the exact origin dual norm; it makes no claim about $\mathbb{E}\langle G_{\text{end}}, C_J G_{\text{end}} \rangle$ away from this projection.

5. **The result is phase-neutral.** No BCC, uniform, modulated, disordered, or other vacuum is assumed or selected. The theorem is a source-action reduction and remains valid if the present candidate functional is later rejected.

7. Evidence map and next decisive calculation

Object or route	Verdict	Exact boundary or next condition
Direct absolute anchor	closed	A7 centering and R-146 give (4.7); no separate representation constant is bounded.
Zero-control origin force	closed	Predictable first-chaos projection and A7/A1 moments give (4.6).
Force equals zero	failed shortcut	Centering is not Euler stationarity; R-153 shows the first variation is not forced to vanish.
Nonzero-feedback Ocone	Clark– rejected transfer	It contains feedback derivatives and is not the fixed-law legal force.
Origin-preserving force	graph closed conditionally	Require the same zero-control law and uniform tangent bound (3.5); full composition also needs exact metric/stabilizer transport.
Stationary low away from zero	open	Need a shifted-base variance/tangent theorem or retain the full direct action.
Complete production owner	open and load-bearing	Prove (1.2) with every forward, adjoint, balanced, low, current, trace, future, forest, and sextic block once.

The next decisive calculation is therefore not another origin-force estimate. It is the generalized eigenvalue or Douglas/Feshbach test for the actual owner-complete production chart at the uniform threshold $-(10/11 - \rho)G_{\text{src}}$. A failure must be retained as an explicit legal production direction; a restricted deterministic chart does not decide it.

8. Devil’s-advocate and code-discipline review

1. **Objection: Gaussian integration by parts differentiates the adapted coefficient. DISMISSED.** In the b -th integration, k_b is $\mathcal{F}_{b-1}$ -measurable and therefore independent of the fresh block ξ_b . Its ξ_b derivative is zero.
2. **Objection: the discrete first-chaos norm equals the whole variance. UPHELD against equality.** It is an orthogonal projection and gives only $\leq$ in general. Equality is claimed only for the completed Brownian predictable-representation setting or a terminal already in the adapted first chaos.
3. **Objection: 11!! assumes independent field components. DISMISSED.** Diagonalize the covariance and use weighted Jensen. Correlated and singular Gaussian components are included.
4. **Objection: a finite numerical trace ceiling proves the analytic all-cutoff bound. UPHELD against that inference.** The theorem uses $T_* < \infty$ from trace convergence. Executable finite-ladder values are regression checks only.
5. **Objection: (5.3) closes T-050. UPHELD against that claim.** The complete production owner lower bound (1.2), its random/nonlinear/revisit scope, and its pathwise/graph compatibility are not proved here.

6. **Objection: the same source square is spent twice. DISMISSED.** The origin force is paid once by Young in (5.5); the owner threshold (5.2) contains the explicit source Hessian once. Forward/legal-reverse/balanced/low must still be evaluated as one Hessian and receive no second source reserve.

The primary executable checks the exact predictable projection on polynomial Gaussian fixtures, the martingale/Bessel form, the twelfth-moment constant, the R-146/R-163 coefficient algebra, records current authority hashes, and checks every scope flag. The non-importing independent executable reconstructs the projection by exact Hermite enumeration and the Gaussian moment by multi-index expansion. Every reported derived constant is recomputed; explicit literals are used only as labelled exact test oracles.

9. Reproduction and Result footer

Dependencies: A1, A7, A11; R-104, R-128, R-141, R-146, R-153, R-163.

Reproduction command: from the repository root, run the primary, independent, and integrated executables below.

```
$py = 'E:\Dev\TECT.venv\Scripts\python.exe'
& $py codes/foundations/a13_classii_predictable_first_chaos_origin_force_
  anchor_free_semiconvexity_reduction.py
& $py codes/foundations/a13_classii_predictable_first_chaos_origin_force_
  anchor_free_semiconvexity_reduction_independent.py
& $py codes/foundations/a13_classii_predictable_first_chaos_origin_force_
  anchor_free_semiconvexity_reduction_verify.py
```

Result ID: A13-CLASSII-PREDICTABLE-FIRST-CHAOS-ORIGIN-FORCE-
ANCHOR-FREE-SEMICONVEXITY-REDUCTION; Ledger ID: R-164.

Precise statement: the legal zero-control projected force is the predictable first-Gaussian-chaos projection of the complete endpoint scalar and has squared source dual norm at most its variance; A7 L2 convergence and A1 point-covariance trace convergence make that variance cutoff-uniform; R-146 removes the direct anchor; a uniform owner margin ρ above $-10/11$ gives $\mu = -1/110 + \rho$, $\epsilon_{\text{v}} = \max(0, 5/11 - \rho/4)$, $\epsilon_6 = 3/20$, and $C = B/\rho$.

Scope: finite temporally faithful Gaussian charts with the A7 zero-control law; direct complete action; fixed floor and side-16 registered A1/A7 sharp-cube/dyadic cutoff sequence.

A force-only graph corollary requires an origin-preserving graph and one uniform tangent bound; full graph composition also requires exact source-metric and stabilizer transport.

Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.

Expected output: primary 50/50, independent 47/47, integrated, PDF, and release checks all PASS.

Falsification gate: any projection, martingale, moment, centering, trace, coefficient, authority, hash, PDF, exploration, public-surface, or release assertion fails.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: no complete production owner lower bound, stationary low elimination away from zero, nonzero-feedback force bound, random/nonlinear/revisit or pathwise conditional semiconvexity, removal, T-050, A13, Nelson, interacting measure, phase/PDE verdict, or Sector-A closure.

Proof complete: false; T-050 closed: false; Sector A closed: false.

Next required action: assemble the actual complete production owner and test $K_{\text{owner}} \geq -(10/11 - \rho)G_{\text{src}}$ with one uniform $\rho > 0$, retaining forward, legal reverse, balanced, low, trace, current, future, forest, and sextic contributions exactly once; record a failed eigenvector as a legal production counterdirection.